Alexander Uamai

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PROFESSIONAL SUMMARY

Mechanical/Mechatronics Engineer who owns electromechanical hardware from concept through production. Currently leading mechanical design, prototyping, and validation of an industrial robotic automation platform — precision assemblies, end-effectors, motion systems, and embedded hardware integration. Hands-on across SolidWorks, GD&T, tolerance analysis, FEA, and DFM/DFA, with a bias toward rapid iteration and manufacturable design. Incoming M.S. in Robotics and Autonomous Systems Engineering, Johns Hopkins University.

PROFESSIONAL EXPERIENCE

Unifiller Systems ULC — Mechatronics Engineering Project Leader (R&D) *Dec 2025 – Present*

Richmond, BC

- Own mechanical design and delivery of an industrial robotic food-automation platform end-to-end, from concept through prototype validation and production support.
- Design precision mechanical assemblies, end-effectors, tooling, and fixtures in SolidWorks, applying GD&T and tolerance analysis to keep designs both high-performing and manufacturable.
- Apply FEA and first-principles analysis to validate mechanical designs before build, then prototype, test, and iterate based on hardware results.
- Integrate motion systems, sensors, actuators, pneumatics, and embedded electronics into a single robotic platform.
- Drive DFM/DFA, supplier selection, and component sourcing, and produce engineering drawings, BOMs, and change documentation to support manufacturing readiness.
- Collaborate daily across mechanical, electrical, software, and manufacturing teams to resolve integration issues and keep hardware development on schedule.

Bethel Systems — Technical & Business Analyst *Dec 2024 – Dec 2025*

- Led engineering product-conceptualization projects for clients, translating early-stage ideas into development plans, including support for patent applications.

Ziva Dynamics — Software Developer Intern *May 2021 – Aug 2021*

- Developed C++ and Python production software with a Qt/PySide2 interface, contributing to Git-based automated testing workflows.

VMAC Air — Mechanical Engineering Intern *Sep 2021 – Dec 2021*

- Designed manufacturing fixtures and production test equipment in SolidWorks to support engineering and manufacturing improvement projects.

Mulgrave School — IT Intern *Jan 2021 – May 2021*

- Built a Raspberry Pi-based remote 3D-printing solution and administered Windows Active Directory.

PROJECTS

- Robotic Automation Platform — Designed end-effectors, tooling, and mechanical hardware for industrial robot motion workflows; led integration, hardware testing, validation, and iterative optimization.
- BLE Embedded Controller — Designed and debugged a BLE-enabled embedded control board using SEGGER J-Link, Tag-Connect, and Code Composer Studio.
- Motor Control System — Developed a PWM motor controller with encoder feedback and embedded firmware on an 8051 platform.

EDUCATION

Johns Hopkins University — M.S., Robotics and Autonomous Systems Engineering *Starts Fall 2026*

Simon Fraser University — B.A.Sc., Mechatronic Systems Engineering *2018 – 2023*

TECHNICAL SKILLS

Mechanical: SolidWorks, Mechanical Design, GD&T, Tolerance Analysis, FEA, DFM, DFA, Engineering Drawings, BOMs, Rapid Prototyping

Robotics & Electromechanical: Industrial Robotics, Motion Control, End-Effector Design, Actuators, Sensors, Pneumatics, Embedded Systems, PCB Integration

Software & Process: C++, Python, MATLAB, Simulink, Qt, Git, GitHub, Linux, Jenkins, Jira, Engineering Change Management